

RELATIONAL PROGRAMMING LANGUAGE GRAMMAR

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GRAMMAR

$session = command^* done$

$command = \left\{ \begin{array}{l} prefixid [identifier] \equiv expression \\ display expression \end{array} \right\}$

$expression = \left\{ \begin{array}{l} [expression infix] application \\ superscription \end{array} \right\}$

$application = \left\{ \begin{array}{l} [application] primary \\ iter [primary \rightarrow primary] \end{array} \right\}$

$superscription = expression \sup \left\{ \begin{array}{l} application \\ + \\ * \end{array} \right\}$

$primary = \left\{ \begin{array}{l} literal \\ prefixid \\ \left[\begin{array}{l} infix \\ \left\{ \begin{array}{l} infix primary \\ primary infix \\ primary \rightarrow primary ; primary \end{array} \right\} \\ (expression [.. expression]) \\ \{ expression [.. expression] \} \\ < primary , \dots > \\ file string \end{array} \right\}$

$infix = infixop [bar]$

$identifier = letter \left[\begin{array}{l} letter \\ digit \end{array} \right]^* prime^*$

$prime = '$

$literal = \left\{ \begin{array}{l} digit^+ [.. digit^+] \\ string \\ true \\ false \end{array} \right\}$

string = “ *char* ”

infixop =

sel | , : cup member nomem !subset subset = -> <- restr ; cl cr cap \
@hat ! cat @ . || \$ red + - times divide != < > <= >=
andsign orsign cart

prefixid = $\left\{ \begin{array}{l} identifier \\ prefixop \end{array} \right\}$

prefixop =

- un cur unc theta size str DELTA inv dom rng mem Lm Rm Mm run lun bun
init term alpha omega ALPHA OMEGA min max mu index select join as sa sa0
rp rpi rsort sort unimg all ssm img curry uncurry PHI Id while epsilon
phi delta PI extend restrict wig not

INPUT FORMS

TABLE 1. Primitive Extensional Operations

Name	Input Form	Publication Form
selection	$t \text{ sel } x$	$t \downarrow x$
relative product	$t \mid u$	$t \mid u$
construction	$t , \text{ bar } u$	$t \bar{,} u$
pair formation	$x : y$	$x : y$
union	$x \text{ cup } y$	$x \cup y$
unit set	$\text{un } x$	$\text{un } x$
currying	$\text{cur } x$	$\text{cur } x$
uncurrying	$\text{unc } x$	$\text{unc } x$
unique element selection	$\text{theta } x$	θx
cardinality	$\text{size } x$	$\text{size } x$
structure	$\text{str } t$	$\text{str } t$
transitive closure	$t \text{ sup } +$	t^+
empty set	empty	$\emptyset$

TABLE 2. Nonprimitive Extensional Operations: Group 1

Name	Input Form	Publication Form
pair list	(x, y)	(x, y)
left pair section	$(x,)$	$(x,)$
right pair section	$(, y)$	$(, y)$
duplication	$\text{DELTA } x$	Δx
membership	$x \text{ member } t$	$x \in t$
nonmembership	$x \text{ nomem } t$	$x \notin t$
improper subset	$s \text{ !subset } t$	$s \subseteq t$
proper subset	$s \text{ subset } t$	$s \subset t$
equality	$s = t$	$s = t$
inverse	$\text{inv } t, t \text{ sup } -1$	$\text{cnv } t, t^{-1}$
domain	$\text{dom } t$	$\text{dom } t$
range	$\text{rng } t$	$\text{rng } t$
members	$\text{mem } t$	$\text{mem } t$
left member	$\text{Lm } (x, t)$	$\text{Lm } (x, t)$
right member	$\text{Rm } (x, t)$	$\text{Rm } (x, t)$
member	$\text{Mm } (x, t)$	$\text{Mm } (x, t)$
right univalent	$\text{run } t$	$\text{run } t$
left univalent	$\text{lun } t$	$\text{lun } t$
bi-univalent	$\text{bun } t$	$\text{bun } t$
initial members	$\text{init } t$	$\text{init } t$
terminal members	$\text{term } t$	$\text{term } t$
reflexive transitive closure	$t \text{ sup } *$	t^*
domain restriction	$p \rightarrow t$	$p \rightarrow t$
range restriction	$t \leftarrow p$	$t \leftarrow p$
restriction	$t \text{ restr } p$	$t \upharpoonright p$

TABLE 3. Nonprimitive Extensional Operations: Group 2

Name	Input Form	Publication Form
first member	alpha t	αt
last member	omega t	ωt
initial sequence	ALPHA t	$A t$
final sequence	OMEGA t	Ωt
ordered union	t ; u	$t ; u$
cons left	x cl t	$x \text{ cl } t$
cons right	t cr x	$t \text{ cr } x$
minimum	min s	$\min s$
maximum	max s	$\max s$
intersection	s cap t	$s \cap t$
set difference	s \ t	$s \setminus t$
subrange set	{m .. n}	$\{m, \dots, n\}$
subrange sequence	(m .. n)	$(m, \dots, n)$
apply functional record	t @hat x	$t \hat{\text{@}} x$
apply functional structure	t ! x	$t ! x$
minimize	mu t	μt
database index	index x t	$\text{index } x t$
database select	select x	$\text{select } x$
database join	join x	$\text{join } x$
array to sequence	as t	$\text{as } t$
sequence to array	sa t	$\text{sa } t$
seq. to zero-origin array	sa0 t	$\text{sa0 } t$
relative product	rp f t	$ f t$
relative product inverse	rpi f t	$ f t$
array catenation	t cat u	$t \text{ cat } u$
relation sort	rsort s	$\text{rsort } s$
sort	sort s	$\text{sort } s$
unit image	unimg t x	$\text{unimg } t x$
all	all t	$\text{all } t$
sequence to matrix	ssm t	$\text{ssm } t$

TABLE 4. Primitive Intensional Operations

Name	Input Form	Publication Form
application	f @ x	$f \text{ @ } x$
image	img f s	$\text{img } f s$
composition	f . g	$f \cdot g, f \cdot g, f \text{ comp } g$
left section	[x+], [x-], ...	$[x+], [x-], \dots$
right section	[+y], [-y], ...	$[+y], [-y], \dots$
paralleling	f g	$f \parallel g$
isomorphism	f \$ t	$f \$ t$
formal application	f @ bar g	$f \bar{\text{@}} g$
functional condition	[p -> f; g]	$[p \rightarrow f; g]$
curry	curry f	$\text{curry } f$
uncurry	uncurry f	$\text{uncurry } f$
filtering	PHI p (d, r)	$\Phi p (d, r)$
iteration	iter [p -> f]	$\text{iter } [p \rightarrow f]$
formalization	+ bar, times bar, ...	$\bar{+}, \bar{\times}$
identity	Id	Id

TABLE 5. Nonprimitive Intensional Operations

Name	Input Form	Publication Form
while loop	while [p, f]	while [p, f]
reduction	f red i	$f \S i$
repeated composition	f sup n	f^n
value of node	upsilon f	$v f$
operate on form	phi f	ϕf
operate on data	delta f	δf
image of structure	PI f	Πf
extension	extend (t, f)	e xtend (t, f)
restriction	restrict (s, f)	restrict (s, f)
formal negation	wig p	$\sim p$

TABLE 6. Miscellaneous Operations

Name	Input Form	Publication Form
sum	x + y	$x + y$
difference	x - y	$x - y$
product	x times y	$x \times y$
quotient	x divide y	x / y
inequality	x != y	$x \neq y$
less	x < y	$x < y$
greater	x > y	$x > y$
less or equal	x <= y	$x \leq y$
greater or equal	x >= y	$x \geq y$
conjunction	x andsign y	$x \wedge y$
disjunction	x orsign y	$x \vee y$
negation	not x	$\neg x$
cartesian product	s cart t	$s \times t$

TABLE 7. Miscellaneous Syntax

Name	Input Form	Publication Form
parentheses	()	()
brackets	[]	[]
braces	{ }	{ }
sequences	<x, y, ... >	<x, y, ... >
identifiers	a, b', total, etc.	a, b', total, etc.
strings	"abcd"	"abcd"
data definition	x == y	$x \equiv y$
function definition	f x == y	$f x \equiv y$
output	display x	display x

EXAMPLES

TABLE 8. Example Program in Publication Form

$F \equiv$ file “OldMaster”	-- F = old file
$U \equiv$ file “Updates”	-- U = update file
$H \equiv$ “Hours”	-- Field name for hours worked
$u \equiv ([H,] . [+]) (F \bar{;} U)$	-- Updating function
$F' \equiv [;] (u \bar{;} F)$	-- New file
display F'	-- Display result

TABLE 9. Example Program in Input Form

F == file “OldMaster”	-- F = old file
U == file “Updates”	-- U = update file
H == “Hours”	-- Field name for hours worked
u == rp ([H,] . [+]) (F , bar U)	-- Updating function
F' == rp [;] (u , bar F)	-- New file
display F'	-- Display result